value of shunt resistance is so adjusted that current $I_g$ produces full scale deflection in the galvanometer. The potential difference across galvanometer is same as the potential difference across shunt resistance.

$$V_{galvanometer} = V_{shunt}$$

$$\Rightarrow I_g R_g = \left(I - I_g\right) S$$

$$S = \frac{I_g}{\left(I - I_g\right)} R_g \text{ Or}$$

$$I_g = \frac{S}{S + R_g} I \Rightarrow I_g \propto I$$

Since, the deflection in the galvanometer is proportional to the current passing through it.

$$\theta = \frac{1}{G} I_g \Rightarrow \theta \propto I_g \Rightarrow \theta \propto I$$

So, the deflection in the galvanometer measures the current I passing through the circuit (ammeter).

Shunt resistance is connected in parallel to galvanometer. Therefore, resistance of ammeter can be determined by computing the effective resistance, which is

$$\frac{1}{R_{eff}} = \frac{1}{R_g} + \frac{1}{S} \Rightarrow R_{eff} = \frac{R_g S}{R_g + S} = R_a$$

Since, the shunt resistance is a very low resistance and the ratio $\frac{S}{R_g}$ is also small. This means, $R_g$ is also small, i.e., the resistance offered by the ammeter is small. So, when we connect ammeter in series, the ammeter will not change the resistance appreciably and also the current in the circuit. For an ideal ammeter, the resistance must be equal to zero. Hence, the reading in ammeter is always lesser than the actual current in the circuit. Let $I_{ideal}$ be current measured from ideal ammeter and $I_{actual}$ be the actual current measured in the circuit by the ammeter. Then, the percentage error in measuring a current through an ammeter is

$$\frac{\Delta I}{I} \times 100\% = \frac{I_{ideal} - I_{actual}}{I_{actual}} \times 100\%$$

### Key points

1. An ammeter is a low resistance instrument and it is always connected in series to the circuit
2. An ideal ammeter has zero resistance
3. In order to increase the range of an ammeter n times, the value of shunt resistance to be connected in parallel is $S = \frac{G}{n-1}$

### Galvanometer to a voltmeter

A voltmeter is an instrument used to measure potential difference across any two points in the electrical circuits. It should not draw any current from the circuit otherwise the value of potential difference to be measured will change.

Voltmeter must have high resistance and when it is connected in parallel, it will not draw appreciable current so that it will indicate the true potential difference.

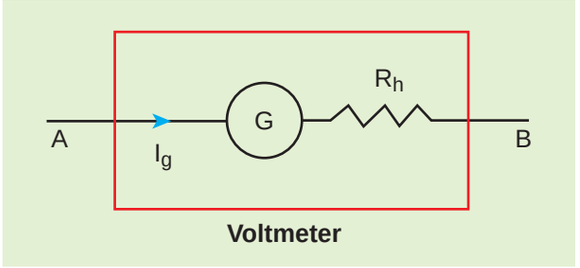



**Figure 3.70** Shunt resistance connected in series



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